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ORIGINAL RESEARCH

Quantitative Computed Tomography Angiography for the Evaluation of Valvular Fibrocalcific Volume in Aortic Stenosis

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ABSTRACT

BACKGROUND Aortic stenosis (AS) is characterized by calcification and fibrosis. The ability to quantify these processes simultaneously has been limited with previous imaging methods.

OBJECTIVES The purpose of this study was to evaluate the aortic valve fibrocalcific volume by computed tomography (CT) angiography in patients with AS, in particular, to assess its reproducibility, association with histology and disease severity, and ability to predict/track progression.

METHODS In 136 patients with AS, fibrocalcific volume was calculated on CT angiograms at baseline and after 1 year. CT attenuation distributions were analyzed using Gaussian-mixture-modeling to derive thresholds for tissue types enabling the quantification of calcific, noncalcific, and fibrocalcific volumes. Scan-rescan reproducibility was assessed and validation provided against histology and in an external cohort.

RESULTS Fibrocalcific volume measurements took 5.8 $\pm$ 1.0 min/scan, demonstrating good correlation with ex vivo valve weight ($r = 0.51$; $P < 0.001$) and excellent scan-rescan reproducibility (mean difference −1%, limits of agreement −4.5% to 2.8%). Baseline fibrocalcific volumes correlated with mean gradient on echocardiography in both male and female participants (rho = 0.64 and 0.69, respectively; both $P < 0.001$) and in the external validation cohort ($n = 66$, rho = 0.58; $P < 0.001$). The relationship was driven principally by calcific volume in men and fibrotic volume in women. After 1 year, fibrocalcific volume increased by 17% and correlated with progression in mean gradient (rho = 0.32; $P = 0.003$). Baseline fibrocalcific volume was the strongest predictor of subsequent mean gradient progression, with a particularly strong association in female patients (rho = 0.75; $P < 0.001$).

CONCLUSIONS The aortic valve fibrocalcific volume provides an anatomic assessment of AS severity that can track disease progression precisely. It correlates with disease severity and hemodynamic progression in both male and female patients. (JACC Cardiovasc Imaging 2024;■:■–■) © 2024 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

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ABBREVIATIONS AND ACRONYMS

AS = aortic stenosis

CT = computed tomography

Aortic stenosis (AS) is the commonest form of valvular heart disease in the Western world and lacks any form of effective medical therapy.[1-3] The disease is characterized by progressive cusp fibrosis and calcification, leading to valve obstruction. The ability to quantify these 2 processes simultaneously has been limited with previous imaging methods.

Echocardiography remains the reference-standard method to assess AS. It assesses valve hemodynamics, primarily using aortic valve velocities, gradients, and derived aortic valve area. However, these echocardiographic measures are discordant in their grading of AS severity in one-quarter to one-third of patients.[4] Moreover, these measures can demonstrate relatively small annualized changes and poor scan-rescan reproducibility.[5]

Aortic valve computed tomography (CT) calcium scoring has emerged as a complementary anatomical assessment of AS severity.[6] It is highly reproducible, independent of flow state, validated against echocardiography, and well suited to the tracking of disease progression.[7] Notable limitations include the inability to assess noncalcific (fibrotic) cusp thickening—an important contributor to stenosis progression, especially in female patients[8]—and anatomical ambiguity caused by the lack of contrast.

Contrast-enhanced cardiac CT angiography has the potential to overcome these limitations.[9] However, previously proposed methods for assessing calcific and noncalcific (fibrotic thickening) aortic valve thickening are time consuming, cumbersome, and unsuitable for clinical practice. Moreover, the ability of CT angiography to predict and track progression of AS and its reproducibility remain unknown.

In the present study, we present a new, rapid, and robust methodology for quantifying the aortic valve fibrocalcific volume in patients with AS. We aimed to investigate the reproducibility of this technique, its association with established hemodynamic markers of AS severity, and its ability to predict and track AS disease progression. Finally, we provide validation of the fibrocalcific volume against histology and in an external cohort of AS patients.

METHODS

STUDY DESIGN AND STUDY POPULATION. This is a post hoc analysis of patients with calcific AS who participated in the SALTIRE 2 (Study Investigating the Effects of Drugs used to Treat Osteoporosis on the Progression of Aortic Stenosis). This was a Medicines and Healthcare products Regulatory Agency-regulated randomized controlled trial of novel drug treatments in AS (NCT02132026) recruiting individuals over 50 years of age with AS (peak aortic jet velocity >2 m/s) from the Edinburgh Heart Centre.[10] SALTIRE 2 was conducted in accordance with the Declaration of Helsinki and approved by the regional ethics committee (Scotland A Research Ethics Committee, 14/SS/0064). All participants provided written informed consent, which explicitly covered permission for post hoc analyses. Participants underwent electrocardiogram-gated, contrast-enhanced cardiac CT angiography at baseline and at 1 year alongside serial echocardiography and noncontrast CT calcium scoring. They received 1 of 3 trial interventions: alendronic acid, denosumab, or matched placebo. All participants were included in this post hoc analysis irrespective of trial intervention, because the primary trial results showed no differences in AS progression between treatment groups and placebo.[10]

A total of 66 patients with AS who underwent CT imaging at Seoul National University Hospital, South Korea, were included as an external validation cohort. These patients were identified retrospectively from the electronic health records, and the study protocol was approved by the local Institutional Review Board. Further details are included in the Supplemental Methods.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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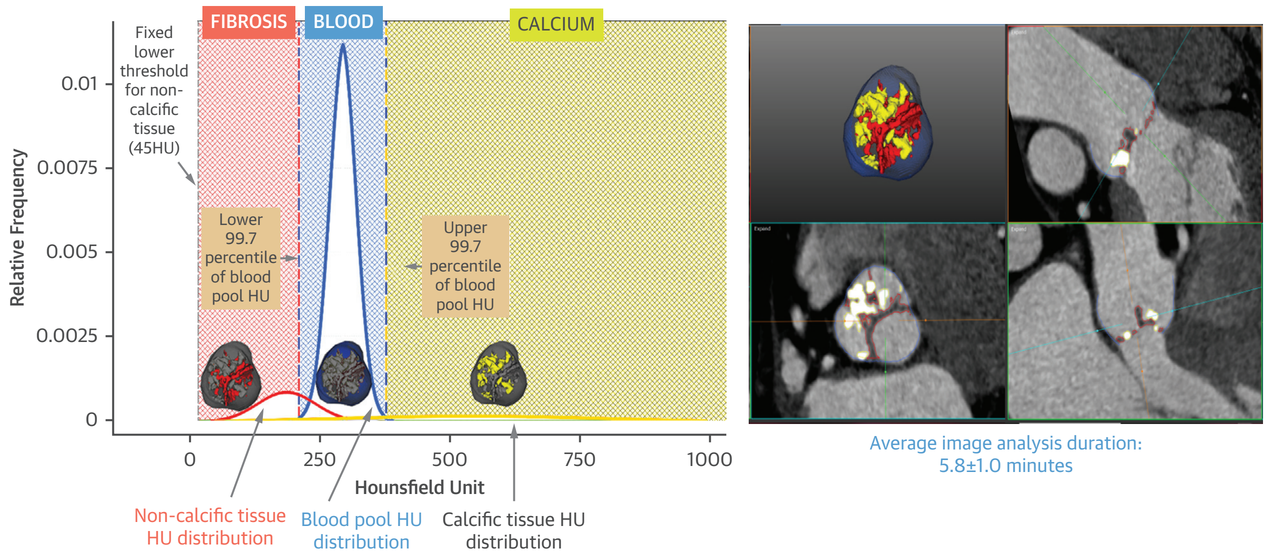
FIGURE 1 Fibrocalcific Volume Assessment



Assessment of fibrocalcific volume from computed tomography angiography (right) and Gaussian mixture model output (left).

STUDY ASSESSMENTS AND DATA COLLECTION. All participants underwent comprehensive clinical assessment and imaging with transthoracic echocardiography, aortic valve Agatston CT calcium score, and contrast-enhanced CT angiography at baseline and 1 year.

Echocardiography. Baseline and 1-year echocardiograms were performed by the same ultrasonographer (A.W.), on the same device (Philips Affinity), in the same accredited department, following the same standardized imaging protocol according to European Association of Cardiovascular Imaging guidelines.[11] AS severity was assessed using the mean aortic valve gradient and peak aortic valve jet velocity.[8,11] Aortic valve area was also calculated using the continuity equation. For all measurements, mean values were taken from 3 measurements when patients were in sinus rhythm and 5 measurements when in atrial fibrillation.

Computed tomography. Baseline and 1-year CT scans were acquired on the same scanner using the same imaging protocol. CT imaging was performed on a 128-detector CT scanner (Biograph mCT, Siemens) using automated tube voltage modulation (CARE Dose 4D, 100-200 kV) with a 0.75-mm slice thickness for contrast-enhanced CT or a tube voltage of 120 kV with a 3-mm slice thickness for noncontrast CT images. Electrocardiogram-gated noncontrast and contrast-enhanced CT angiography were acquired in diastole and in held expiration. Patients were administered beta-blockade for heart rate control when their resting heart rate was >65 beats/min and there were no contraindications. Aortic valve CT Agatston calcium score was assessed by an experienced operator (T.P.) using Vitrea software (Vitrea Advanced, Vital Images).[6]

GAUSSIAN MIXTURE MODEL FOR THE EVALUATION OF FIBROCALCIFIC VOLUME. The computation of calcific and noncalcific volumes on contrast-enhanced CT angiograms was assessed using the valve biopsy module of Autoplaque software (version 2.5, Cedars-Sinai Medical Center) on a diastolic phase (usually 70% R-R interval) so that the heart valve leaflets were closed. Tissues with different pixel intensity (eg, blood pool, calcific, and noncalcific tissue) within an overall population were obtained from semiautomated segmentation of the aortic valve complex (Figure 1). Gaussian mixture modeling was applied to the histogram of attenuation values derived from this data, and a probabilistic model then derived thresholds (Hounsfield Units [HU]) to define each tissue type. This algorithm has been applied for coronary plaque characterization from CT angiography,[12] and also to magnetic resonance imaging of the brain and CT imaging of the liver for tissue characterization.[13,14]

Scans with motion artefact or inadequate contrast opacification of the lumen (<150 HU) around the aortic valve were identified by visual assessment and then excluded. After orientating the CT scan into the axis of the aortic valve annular plane according to standardized methods,[15] a volume of interest incorporating the entire aortic valve (from the annular plane to the origin of the right coronary artery) was defined by adjusting the appropriate radius and height. The region of interest was automatically divided by the software into 5 slices including the aortic valve and its contour. For each slice, manual adjustment of the automated valve contour was performed to exclude calcific and noncalcific thickening arising from external structures surrounding the aortic valve, eg, the aortic wall, coronary arteries, mitral valve annulus, left ventricular outflow tract, and aortic annulus.

Using the Gaussian mixture model, the software estimated the CT attenuation (HU) distribution of the 3 populations (blood pool, noncalcific, and calcific tissue) within the aortic valve volume of interest (**Figure 1**). For noncalcific tissue, a fixed lower threshold of 45 HU was employed to exclude artefacts (eg, photon starvation adjacent to dense calcification), while an upper threshold was set as the 0.3 percentile of the blood pool CT attenuation. For calcific tissue, the lower threshold was set as the 99.7 percentile of blood pool CT attenuation, and all tissue above this threshold was considered to be calcium. To account for differences in valve size, calcific and noncalcific volumes were indexed to the annulus area as described previously.[9] The fibrocalcific volume was defined as the sum of the calcific and noncalcific volumes and the fibrocalcific ratio was calculated by dividing the noncalcific volume by the calcific volume. Analysis was performed by experienced reporters, and the time taken to perform the analysis was measured, including loading of the scans onto the software, selecting and refining the contours for the volume of interest, measuring the annulus area, checking the derived fibrotic and calcific volumes, and exporting the data. A step-by-step protocol is available in the Supplemental Methods.

Repeatability and reproducibility. To assess intraobserver repeatability, aortic valve calcific and noncalcific volumes were measured on 20 anonymized contrast-enhanced CT angiograms by the same observer (M.L.) on 2 occasions, at least 4 weeks apart in random order to minimize recall bias. To assess interobserver repeatability, 20 scans were independently analyzed by 2 trained observers (M.L. and S.J.) 4 weeks apart. Scan-rescan reproducibility was assessed in 13 patients who had contrast CT at baseline and a repeat scan using the same protocol on the same scanner within 4 weeks. Image analysis of the repeated CT scans was performed by a trained operator (M.L.) more than 2 weeks apart to minimize recall bias.

Histological validation. Aortic valves from 41 patients undergoing valve replacement surgery at Institut Universitaire de Cardiologie et de Pneumologie de Québec (Canada) were included in this cohort. Semi-quantitative assessment of valve fibrosis (1: mild, 2: moderate, and 3: severe) and calcification (Warren-Yong score—1: absent, 2: mild valve thickening and early nodular calcification, 3: moderate thickening with many calcified nodules, 4: severe thickening with many calcified nodules) was performed. Further details are included in the Supplemental Methods.

STATISTICAL ANALYSIS. All statistical analyses were performed using R studio (version 1.1.1093) or SPSS (version 26). Continuous normally distributed variables were expressed as mean $\pm$ SD of the mean or median (Q1, Q3) when not normally distributed. Normality of continuous measures was assessed using Shapiro-Wilk's test. Parametric (unpaired Student's *t*-test) or nonparametric (Mann-Whitney U) tests were used for independent variables as appropriate. Categorical data were presented as number (percentage) and compared using the chi-square test. Intraobserver and interobserver repeatability and scan-rescan reproducibility were assessed using Bland-Altman analysis and intraclass correlation coefficient (2-way agreement model). Correlations between continuous variables were assessed with linear regression analysis and either Pearson's or Spearman's Rho, subject to normality. Rates of progression were calculated using the difference between 1 year and baseline for all echocardiographic and CT assessments, and progression analysis was limited to the participants whose baseline and 1-year scans were of sufficient quality. Multivariable linear regression analysis was performed to investigate baseline predictors of subsequent AS progression measured with echocardiography and covariates such as age, sex, hypertension, hypercholesterolemia, diabetes, and serum creatinine. Statistical significance was defined as 2-sided value of $P < 0.05$.

Based upon the scan-rescan repeatability and progression data from this study, we calculated putative group sizes that would be required to detect treatment effects (30%, 20%, and 10% reductions in the annualized progression) of a new therapy using the fibrocalcific volume and an alpha of 0.05 at both 80% and 90% power.

TABLE 1 Baseline Demographics and Imaging Parameters of Study Population

	Overall (N = 136)	Male Participants (n = 107)	Female Participants (n = 19)	*P* Value
Demographics				
Age, y	72 (68-76)	72 (68-76)	72 (67-77)	0.73
Hypertension	104 (76)	82 (77)	22 (76)	0.93
Diabetes mellitus	30 (22)	23 (21)	7 (24)	0.76
Hypercholesterolemia	80 (59)	64 (60)	16 (55)	0.65
Angina	37 (27)	28 (26)	9 (31)	0.61
Previous myocardial infarction	17 (13)	16 (15)	1 (3)	0.09
Previous coronary artery bypass grafting	15 (11)	13 (12)	2 (7)	0.42
Imaging parameters at baseline				
Echocardiography				
Aortic valve peak velocity, m/s	3.4 (3.0-3.8)	3.5 (3.0-3.9)	3.2 (2.8-3.5)	0.02
Aortic valve mean gradient, mm Hg	23.8 (18.3-31.8)	25.9 (18.9-33.0)	21.4 (16.9-27.6)	0.07
Aortic valve area, cm^2	1.08 (0.88-1.26)	1.02 (0.88-1.26)	1.13 (1.00-1.28)	0.61
Stroke volume index, mL/m^2	42.0 (37.1-48.0)	41.5 (35.8-47.0)	44.9 (44.6-49.9)	0.07
Mild AS	31 (22)	22 (20.5)	9 (31)	0.34
Moderate AS	81 (60)	63 (59)	18 (62)	0.92
Severe AS	24 (18)	22 (20.5)	2 (7)	0.15
CT				
Agatston calcium Score	1,141 (649.7-2,153.3)	1,268 (760.0-2,338.0)	460 (205.5-1,110.0)	<0.0001
Calcific volume, $\mu L/cm^2$	51 (24.8-101.0)	64 (36.6-122.6)	16 (7.6-41.4)	<0.0001
Noncalcific volume, $\mu L/cm^2$	156 (113.6-194.8)	156 (114.3-194.7)	163 (96.9-200.3)	0.70
Fibrocalcific volume, $\mu L/cm^2$	223 (172.0-271.4)	235 (189.1-274.7)	182 (129.3-238.9)	0.006
Fibrocalcific ratio	2.7 (1.2-6.0)	2.1 (1.1-4.5)	6.4 (2.9-16.0)	<0.0001

Values are n (%) or median (Q1-Q3).
AS = aortic stenosis; CT = computed tomography.

RESULTS

PATIENT CHARACTERISTICS. Of the 155 patients included in the clinical trial, 19 patients (12%) were excluded because of inadequate image quality on visual analysis (motion artefact in 6 scans; poor contrast opacification in 13 scans), leaving 136 participants in the baseline study population. Participants had a median age of 72 years (Q1-Q3: 68-76 years), most were men (79%), and a minority (8%) had a bicuspid aortic valve (Table 1). All patients had a left ventricular ejection fraction >50%. At baseline, 24 (18%) patients had severe, 81 (60%) moderate, and 31 (22%) mild AS.

FIBROCALCIFIC VOLUME MEASUREMENTS. The mean duration of the analysis was 5.8 ± 1.0 min/scan. Measurements of the aortic valve fibrocalcific volumes demonstrated excellent repeatability and rescan reproducibility with no fixed or proportional biases and tight limits of agreement (Table 2, Figure 2). With respect to scan-rescan reproducibility, the mean difference was −1%, and limits of agreement −4.5% to 2.8% (Table 2, Figure 2). Some examples of aortic valve fibrocalcific volume evaluation are shown in Figure 3.

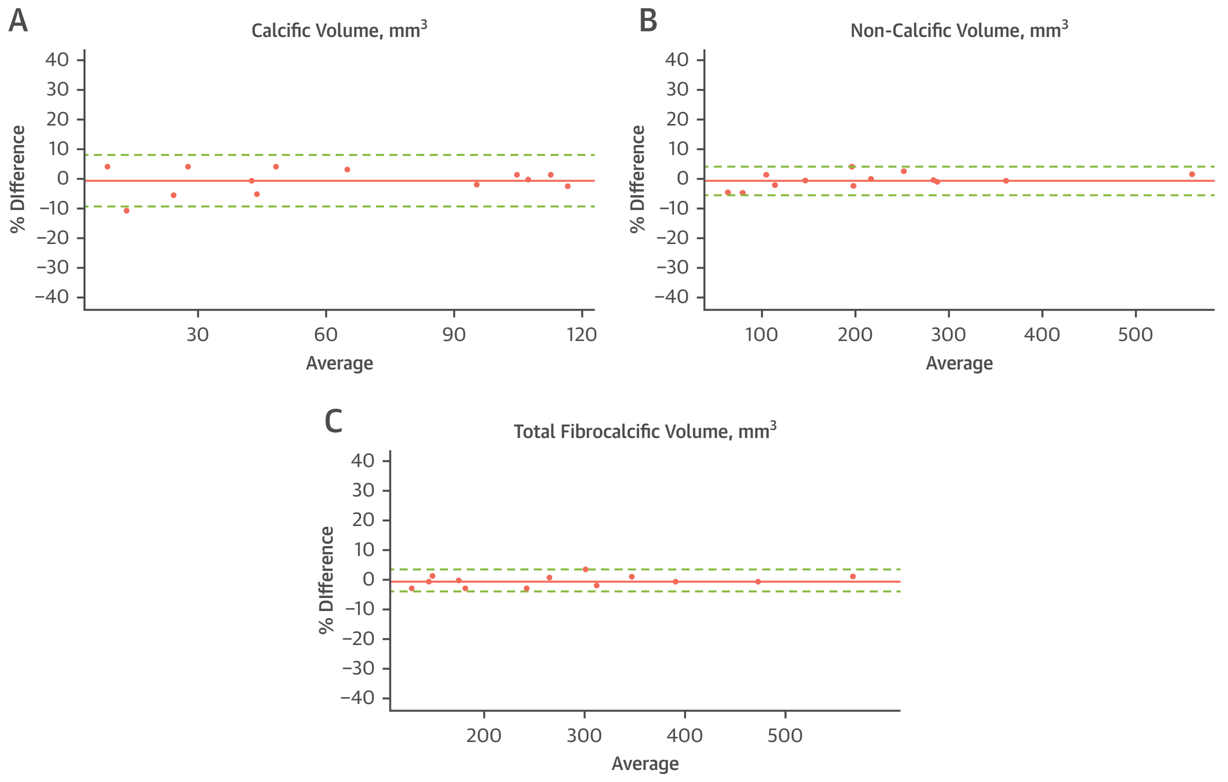
BASELINE ASSESSMENTS OF AS SEVERITY. The median calcific, noncalcific, and fibrocalcific volumes for the study population were 51 $\mu L/cm^2$ (Q1-Q3: 26-100 $\mu L/cm^2$), 156 $\mu L/cm^2$ (Q1-Q3: 114-195 $\mu L/cm^2$), and 223 $\mu L/cm^2$ (Q1-Q3: 172-271 $\mu L/cm^2$), respectively (Table 3). Baseline calcific volumes correlated very strongly with CT Agatston calcium scores of the aortic valve (rho = 0.90; $P < 0.0001$) but were not associated with baseline noncalcific volumes (rho = −0.052; $P = 0.54$).

Fibrocalcific volumes demonstrated a close association with baseline echocardiographic assessments of AS severity (mean gradient rho = 0.66; $P < 0.0001$) (Supplemental Figure 1). The correlation between the fibrocalcific volume and mean gradient was similar in male (rho = 0.64) and female patients (rho = 0.69) (both $P < 0.0001$). However, the strength of this correlation was mainly driven by the calcific volume (rho = 0.58) in male patients and by noncalcific volume (rho = 0.49) in female patients (Supplemental Table 1). Consistent with this, the median fibrocalcific ratio for male participants was 2.1 (Q1-Q3: 1.1-4.5) compared with 6.4 (Q1-Q3: 3.5-16) for female participants ($P < 0.0001$). In female participants, the fibrocalcific volume demonstrated a closer association with the mean gradient on echocardiography than the

TABLE 2 Repeatability and Reproducibility Measurements for the Fibrocalcific Volume

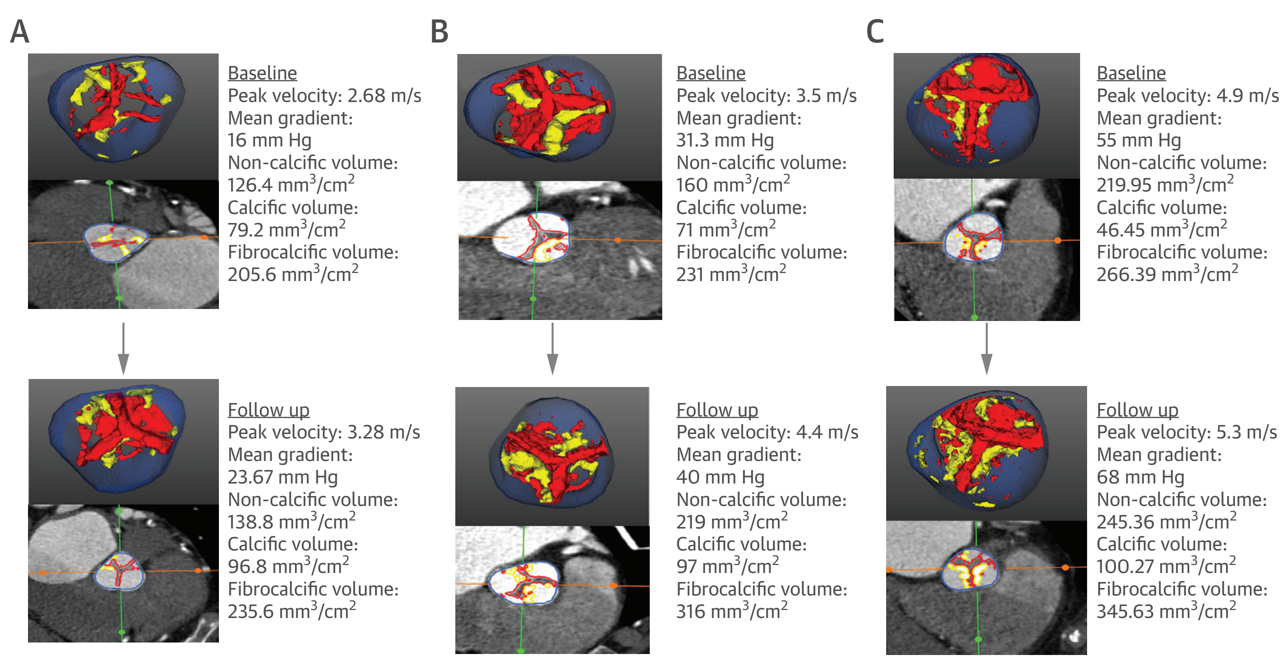
	Mean Difference, %	Lower Limit of Agreement, %	Upper Limit of Agreement, %	Absolute Values Mean Difference (Limits of Agreement), μL/cm^2	Intraclass Correlation (95% CIs)
Intraobserver repeatability					
Calcific volume	−0.3	−4.4	3.9	−0.3 (−2.9 to 2.3)	1.0 (0.99-1.00)
Noncalcific volume	0.2	−3.1	3.5	0.4 (−5.5 to 6.4)	0.99 (0.99-1.00)
Fibrocalcific volume	0.2	−2.1	2.5	0.2 (−6.1 to 6.4)	0.99 (0.99-1.00)
Interobserver repeatability					
Calcific volume	5.0	−7.9	17.9	3.5 (−4.5 to 11.5)	0.98 (0.88-0.99)
Noncalcific volume	1.2	−9.8	12.2	2.6 (−16.2 to 21.4)	0.95 (0.83-0.98)
Fibrocalcific volume	2.2	−6.4	10.8	6.1 (−16.8 to 29.0)	0.97 (0.87-0.99)
Scan-rescan reproducibility					
Calcific volume	−0.6	−9.4	8.2	−0.1 (−3.5 to 3.3)	0.99 (0.99-1.00)
Noncalcific volume	−1.3	−6.1	3.6	−1.4 (−9.4 to 6.6)	1.0 (0.99-1.00)
Fibrocalcific volume	−0.9	−4.5	2.8	−1.5 (−10.6 to 7.6)	0.99 (0.99-1.00)

FIGURE 2 Bland Altman Plots for Scan-Rescan Reproducibility



Calcific volume (A), noncalcific volume (B), and fibrocalcific volume (C). The dotted red lines represent 95% CI limits of agreement, and the solid black line represents mean difference.

FIGURE 3 **Disease Progression Assessment Using Fibrocalcific Volume**



Examples of disease progression as assessed using the fibrocalcific score, comparing computed tomography images at baseline (top) and after 1 year of follow-up (bottom). Participant with mild aortic stenosis (AS) at baseline (A), participant with moderate AS at baseline (B), and participant with severe AS at baseline (C).

aortic valve CT calcium score (rho = 0.69 vs rho = 0.47, respectively).

Similar results were observed when comparing contemporaneous echocardiogram and CT assessments of AS severity acquired at the 1-year time point (Supplemental Table 2).

DISEASE PROGRESSION. At 1-year follow-up, 90 participants from the main clinical cohort had CT angiograms of sufficient image quality for analysis: 27 participants dropped out of the study and 19 scans had inadequate image quality (motion artefact on 5 scans; poor contrast opacification on 14 scans). There was no effect of the trial interventions on the progression of calcific, noncalcific, and fibrocalcific volumes after 1 year (Supplemental Table 3).

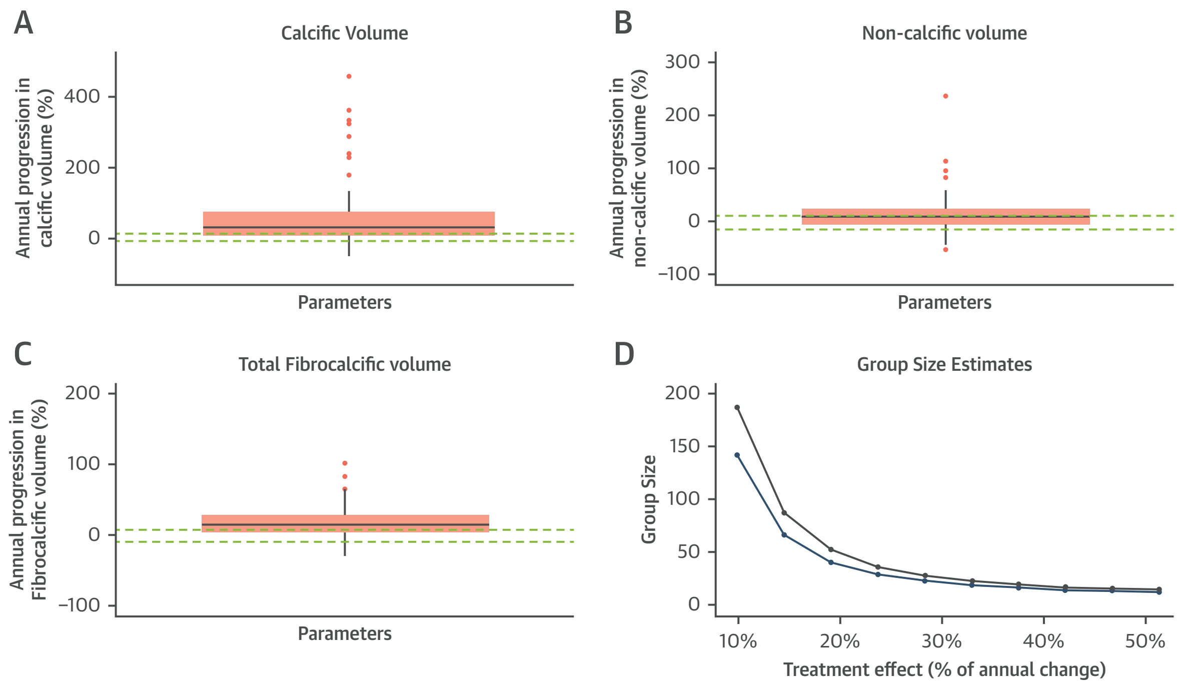
At 1 year, the indexed fibrocalcific volume increased by 17% (absolute change: 37.1 ± 37.9 μL/cm²) (**Table 3**, **Figure 4**). This change correlated with echocardiographic assessments of hemodynamic disease progression (change in aortic valve mean gradient rho = 0.32; $P = 0.003$) (Supplemental Table 4). Baseline fibrocalcific volume also predicted subsequent AS progression assessed by echocardiography (change in mean gradient rho = 0.39; $P < 0.0001$),

TABLE 3 **Disease Progression Measured With Contrast CT Angiography, CT Calcium Scoring, and Echocardiography**

	Baseline	1-y Follow-Up	Percentage Change, %	Absolute Change	Absolute Change	*P* Value
Indexed noncalcific volume, μL/cm²	156 (114-195)	158 (115-205)	8	12 (−7 to 35)	13.1 ± 37.2	0.003
Indexed calcific volume, μL/cm²	51 (25-101)	79 (42-139)	37	16 (7-38)	25.0 ± 31.4	<0.0001
Indexed fibrocalcific volume, μL/cm²	223 (172-271)	244 (198-305)	17	32 (7-60)	37.1 ± 37.9	<0.0001
Agatston calcium score	1,141 (650-2,153)	1,319 (741-2,398)	22	214 (81-401)	304.1 ± 363.7	<0.0001
Aortic valve peak velocity, m/s	3.37 (3.01-3.87)	3.65 (3.09-4.12)	8	0.27 (0.11-0.49)	0.28 ± 0.27	<0.0001
Aortic valve mean gradient, mm Hg	23.8 (18.3-31.8)	27.0 (18.9-35.4)	18	3.2 (−0.3 to 7.6)	4.2 ± 7.2	<0.0001
Aortic valve area, cm²	1.08 (0.88-1.24)	0.99 (0.80-1.17)	9	−0.10 (−0.17 to −0.04)	−0.12 ± 0.12	<0.0001

Values are median (Q1-Q3) or mean ± SD. *P* value was measured for the change from baseline to 1-year follow-up values for different parameters.
Abbreviation as in **Table 1**.

FIGURE 4 Annualized Progression and Sample Size Calculation Using Fibrocalcific Volume



Annualized progression for calcific volume (A), noncalcific volume (B), and fibrocalcific volume (C). Dotted red lines represent expected noise (limits of agreement for scan-rescan reproducibility), which is consistently less than the annualized progression, indicating the ability of these measurements to assess disease progression. (D) Graph representing sample size estimates for putative studies using the fibrocalcific volume to investigate the effects of novel therapies on AS progression. Sample sizes are provided for different treatment effects at both 90% (green line) and 80% (blue line) power (alpha = 0.05). Abbreviation as in Figure 3.

outperforming the baseline Agatston calcium score (Table 4). The association between the baseline fibrocalcific volume and progression of the mean gradient was particularly strong in female participants ($r = 0.75$; $P < 0.001$). On multivariable analysis, baseline fibrocalcific volume was the strongest predictor of subsequent disease progression (beta coefficient = 0.27; $P = 0.02$ on multivariable analysis). Other covariates are included in Supplemental Table 5.

HISTOLOGICAL VALIDATION. Valve weight was available in 41 patients. There was a good correlation between fibrocalcific volume and valve weight ($r = 0.51$; $P < 0.001$) (Supplemental Figure 2). This correlation was stronger than the association between valve weight and hemodynamic assessments of valve severity from echocardiography (mean gradient $r = 0.43$; $P = 0.004$; peak aortic jet velocity $r = 0.36$; $P = 0.04$). Histological examination confirmed the presence of valve fibrosis that spatially correlated with areas of noncalcific cusp thickening on CT (Supplemental Figure 3). Furthermore, indexed noncalcific volumes on CT were higher in patients with a higher fibrosis semiquantitative score on histology and similarly indexed calcific volumes on CT were higher in patients with higher Warren-Yong scores for calcification on histology (Supplemental Figure 2).

EXTERNAL CLINICAL VALIDATION. A total of 66 patients were included in the external clinical validation cohort. Baseline demographic and imaging parameters of the validation cohort are summarized in Supplemental Table 6. Similar to the main cohort, fibrocalcific volumes demonstrated a good correlation with baseline echocardiographic assessments of AS severity (mean gradient: rho = 0.58; $P < 0.001$, aortic valve peak velocity: rho = 0.57; $P < 0.001$ and aortic valve area: rho = −0.42; $P = 0.02$) (Supplemental Figure 4).

GROUP SIZE ESTIMATION. Group size estimates in putative studies using the fibrocalcific volume to detect a treatment effect of a new therapy were estimated at 21, 46, and 170 participants, based upon 30%, 20%, and 10% reductions in the

TABLE 4 Correlations Between Baseline CT Parameters and Subsequent AS Progression as Assessed by Echocardiography

	Change in Aortic Valve Mean Gradient, mm Hg			Change in Aortic Valve Peak Velocity, m/s		
	Overall Population	Male Participants	Female Participants	Overall Population	Male Participants	Female Participants
Baseline indexed noncalcific volume, μL/cm^2	rho = 0.22 P = 0.03	rho = 0.13 P = 0.26	rho = 0.59 P = 0.015	rho = 0.14 P = 0.19	rho = 0.07 P = 0.55	rho = 0.44 P = 0.078
Baseline indexed calcific volume, μL/cm^2	rho = 0.32 P = 0.003	rho = 0.27 P = 0.023	rho = 0.38 P = 0.14	rho = 0.24 P = 0.023	rho = 0.20 P = 0.08	rho = 0.37 P = 0.13
Baseline indexed fibrocalcific volume, μL/cm^2	rho = 0.39 P < 0.0001	rho = 0.32 P = 0.006	rho = 0.75 P < 0.001	rho = 0.29 P = 0.006	rho = 0.24 P = 0.04	rho = 0.63 P = 0.007
Baseline Agatston calcium score	rho = 0.34 P < 0.001	rho = 0.28 P = 0.016	rho = 0.52 P = 0.035	rho = 0.26 P = 0.012	rho = 0.20 P = 0.08	rho = 0.55 P = 0.021

Abbreviations as in Table 1.

annualized progression, respectively (alpha = 0.05, power = 90%) (Figure 4).

DISCUSSION

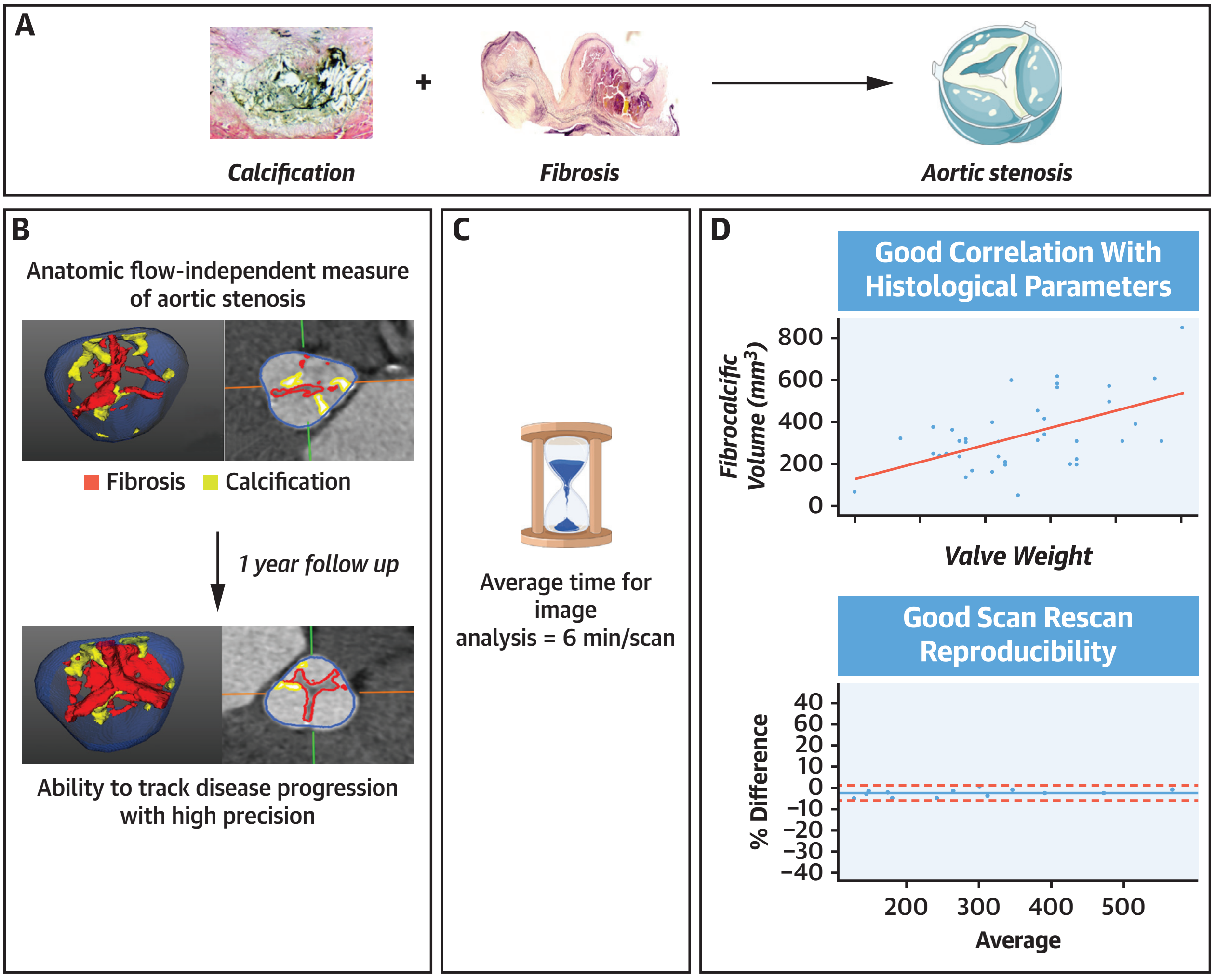
We present a rapid and reproducible assessment of AS severity based on CT angiography: the aortic valve fibrocalcific volume (Central Illustration). This anatomic assessment describes the total burden of disease in the aortic valve, measuring both calcific and fibrotic thickening, and demonstrates a close correlation with hemodynamic AS severity in both male and female patients. The fibrocalcific volume also reveals important sex-specific differences in the pathology of AS, confirming the relative importance of fibrosis in female patients, and demonstrates strong capacity to both predict and track disease progression in patients with AS.

The aortic valve fibrocalcific volume holds several conceptual advantages in the assessment of AS. It provides an anatomic flow-independent measure that is not subject to fluctuations in loading conditions or stroke volume and quantifies the overall aortic valve disease burden. Unlike CT calcium scoring, it does not ignore the contribution of fibrosis to progressive aortic valve stenosis. Fibrosis is the dominant driver of disease progression for a substantial proportion of patients with AS, and this is of particular relevance to female patients, where valve fibrosis makes a larger contribution to the pathology of AS than in male patients.[16] Indeed, in this study, whilst similar correlations between fibrocalcific volume and disease severity were observed in both sexes, this correlation was mainly driven by the calcific volume in male patients and noncalcific volume in female patients. Moreover, the fibrocalcific volume appeared more closely associated with disease progression in female patients than measures focusing on calcium. Finally, CT angiography protocol provides excellent spatial resolution making it much easier to differentiate calcific and noncalcific thickening of the aortic valve cusps from that originating from adjacent structures, another limitation of CT calcium scoring.

Previous studies using CT angiography have described alternative methods to measure fibrosis based upon differences in CT attenuation.[17] However, consistent and robust attenuation thresholds to define calcific and noncalcific thickening have been elusive, principally because of the wide variation in contrast concentrations in the blood pool. Later iterations improved upon this challenge but were cumbersome and time consuming, making them impractical for clinical use.[9] We here present a novel approach that, for the first time, uses a Gaussian mixture model[13] to derive robust thresholds for calcific and noncalcific valve thickening. These thresholds are adaptive, being based upon the relative distributions of CT attenuation in the aortic valve and surrounding blood pool observed on the scan being interrogated. This approach is built into semi-automated software, so that once the operator has identified the aortic valve plane and the volume of interest around the valve, the software automatically calculates calcific, noncalcific, and fibrocalcific volumes. This reduces both human error and analysis time, and results in excellent intraobserver and interobserver repeatability as well as excellent scan-rescan reproducibility. Indeed, no fixed or proportional biases and narrow limits of agreement for this approach were observed. The Gaussian mixture model approach therefore appears to be a robust and rapid method of defining regions of calcific and noncalcific thickening in the aortic valve that we have here validated both histologically and in an external clinical validation cohort.

We have previously described the measurement of the fibrocalcific volume and undertaken histological validation to demonstrate the validity of our

CENTRAL ILLUSTRATION Quantitative Computed Tomography Angiography for the Evaluation of Valvular Fibrocalcific Volume in Aortic Stenosis



Lembo M, et al. JACC Cardiovasc Imaging. 2024;■(■):■-■.

(A) Pathophysiology of aortic stenosis—calcification and fibrosis. (B) Fibrocalcific volume provides anatomic, flow independent assessment of aortic stenosis and has the ability to track disease progression with high precision (areas in red represent noncalcific valve thickening and areas in yellow represent calcific thickening). (C) Fibrocalcific volume measurement is quick and image analysis takes approximately 6 min/scan. (D) Fibrocalcific volume correlates with histological parameters and has excellent scan-rescan reproducibility.

approach. Those previous studies demonstrated that the noncalcific volumes principally relate to areas of valve fibrosis and demonstrated that the fibrocalcific volume had the strongest correlation with disease severity.[9] In the current study, we used a more rapid and robust image analysis technique with similar histological validation and again demonstrated that the fibrocalcific volume correlates well with hemodynamic disease severity. Indeed, the fibrocalcific volume correlated very well with baseline hemodynamic assessments of AS on echocardiography with closer associations than those observed for the calcific and noncalcific volumes in isolation. In female patients, the fibrocalcific score demonstrated a much closer association with echocardiographic hemodynamic assessments than Agatston CT calcium scoring, again highlighting the importance of considering valve fibrosis in female patients.

For the first time, we have here investigated the associations between the fibrocalcific volume and AS disease progression. We demonstrate the ability of this technique to track disease progression with time and its unique ability to characterize changes in both the calcific and noncalcific valve burden. The fibrocalcific volume increased by 17% after 1 year, demonstrating a close association with hemodynamic progression assessed on echocardiography during the same period. Furthermore, baseline fibrocalcific volume outperformed baseline Agatston calcium score and was able to predict subsequent disease progression. Fibrocalcific volume was a particularly strong predictor of disease progression in female patients, emphasizing the role of fibrosis in AS in female patients.

Combined with its excellent scan-rescan reproducibility, these data indicate that the fibrocalcific volume is well suited to track disease progression in patients with AS, resulting in several important implications. First, it addresses the differences in potential causes of disease progression. Both valve fibrosis and valve calcification are important in the pathogenesis and progression of AS, and this approach will provide a measure of the impact of potential treatment interventions on both valvular fibrosis and calcification. Second, it addresses a potential sex bias in the assessment of AS. Female patients develop less valvular calcification than male patients and appear to have a greater contribution of fibrosis.[16,18] This is consistent with the greater correlation of hemodynamic disease progression with the fibrocalcific volume than isolated measures of calcification, something which was less apparent in men. Third, this approach provides a more global assessment of disease burden in the aortic valve that is arguably closer to the actual pathological disease process than echocardiography, which predominantly assesses the functional consequences of aortic valve disease. Thus, anatomical assessments of disease burden (the fibrocalcific volume) may represent a more sensitive and reproducible method of detecting effects of treatment interventions than the consequences for valve function (mean aortic valve gradient), although we acknowledge that the latter is currently more clinically important for patient management.[19] Indeed, given its favorable reproducibility and progression data, anticipated sample sizes for studies using the fibrocalcific volume to investigate whether novel treatments impact disease progression are approximately 10-fold smaller than for similar studies using echocardiography.[8] We believe that our findings therefore have important implications for the future design of trials evaluating novel therapies for AS and note that the upcoming international Lp(a) CAVS FRONTIERS (A Multicenter Trial Assessing the Impact of Lipoprotein(a) Lowering With Pelacarsen [TQJ230] on the Progression of Calcific Aortic Valve Stenosis; NCT05646381) randomized controlled trial has incorporated progression in the aortic valve fibrocalcific volume as an endpoint.

STUDY LIMITATIONS. Diagnostic image quality and adequate contrast opacification are required for the measurement of the fibrocalcific volume. We were unable to analyze approximately 10% of CT angiograms because of motion artifact and poor image quality, highlighting the importance of rigorous imaging protocols. The generalizability of our findings needs confirmation. In particular, female participants and those with bicuspid aortic valves formed modest-sized subgroups in our cohort, and further data from these populations will be needed. Our preliminary data were obtained from a single center, and although we observed similar findings in an external validation cohort, the general clinical utility of this technique remains uncertain with larger multicenter studies required. Moreover, we need to establish prospective thresholds for the fibrocalcific volume that can be used to identify and to assess disease severity in patients with AS. A large international multicenter study is currently underway to derive and to validate these thresholds and to investigate their ability to predict adverse clinical outcomes. Furthermore, a detailed and dedicated histological follow-up study would provide further validation for this technique. Finally, the software used to assess fibrocalcific volume in this study is currently not widely available.

CONCLUSIONS

The aortic valve fibrocalcific volume provides anatomic evaluation of AS severity that measures both calcific and fibrotic thickening of the aortic valve. Our novel approach took an average of 6 minutes to perform. It is both quick to learn and reproducible, and it demonstrates a close association with other markers of AS severity in both male and female patients. Fibrocalcific volume measurements have major potential for both the prediction and tracking of disease progression and the assessment of treatment responses to novel therapies.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: AS is characterized by progressive cusp fibrosis and calcification. Current imaging technologies either have high variability in repeated measurements or are unable to measure fibrotic thickening of the aortic valve cusps.

TRANSLATIONAL OUTLOOK: Fibrocalcific volume is a novel CT-derived measurement that provides assessment of both calcific and fibrotic thickening in AS. Fibrocalcific volume measurement is reproducible, can be performed in 6 minutes, and demonstrates a close association with other markers of AS severity. Fibrocalcific volume has major potential for both the prediction and tracking of disease progression as well as the assessment of treatment responses to novel therapies. It is being investigated as a clinical endpoint in the Lp(a) FRONTIERS CAVS study (NCT05646381).

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KEY WORDS aortic valve stenosis, contrast enhanced CT, fibrocalcific volume, gaussian mixture model

APPENDIX For an expanded Methods section and supplemental tables, figures, and references, please see the online version of this paper.